Time as Projection: The Klein-Gordon Spectrum on $S^3 \times \mathbb{R}$ and the Algebraic / Observation Split

GeoVac Project

May 2026

Abstract

The bare relativistic Klein-Gordon spectrum on $S^3 \times \mathbb{R}$ is π-free in the algebraic-extension ring $\mathbb{Q}[\sqrt{d_1}, \sqrt{d_2}, \ldots]$ with d_i positive square-free integers, for every rational mass-squared parameter. We verify this symbolically for $n \in [1, 50]$ across the panel $m^2 \in \{0, 1, 1/4, 2\}$ (200 cases, zero transcendentals). Mass is therefore a parameter projection that preserves the bare-graph algebraic ring; it does not introduce π.

π enters the spectrum at exactly one step: temporal compactification. On $S^3 \times S^1_\beta$ the temporal direction acquires Matsubara modes $\omega_k^t = 2\pi k/\beta$, and the first π-bearing eigenvalue is the $(n{=}0, k{=}1)$ mode with $\omega^2 = 4\pi^2$. We exhibit the before/after explicitly: every $(n, 0)$ eigenvalue is π-free; every $(n, k{\neq}0)$ eigenvalue contains $2\pi/\beta$.

The conformally coupled massless scalar Casimir energy on unit S^3 (spatial only, no temporal compactification) evaluates to the exact rational $E_{\mathrm{Cas}} = 1/240$, with no transcendental content at any step of the ζ-regularized derivation. The Stefan-Boltzmann constant $\pi^2/90$ appears in the high-temperature expansion of the free-energy density on $S^3 \times S^1_\beta$ only via the Matsubara sum, exactly through the mechanism of the spectral π-injection.

We propose a refinement of Paper 34's projection dictionary: the implicit "scalar parameter projection" should be split into (i) a *rest-mass projection* (introduces m, dimension mass, transcendental signature trivial, ring-preserving for $m^2 \in \mathbb{Q}$) and (ii) an *observation / temporal-window projection* (introduces β, dimension time, transcendental signature $2\pi \cdot \mathbb{Q}$ per Matsubara mode; Stefan-Boltzmann $\pi^2/90$ in the high-T limit). These are categorically distinct; lumping them as a single "parameter row" under-resolves the transcendental axis of the dictionary.

The structural reading is sharper than Paper 34's layer description: *a GeoVac observable contains π if and only if its evaluation includes a continuous integration over a temporal or spectral parameter that has been promoted from the discrete graph spectrum.* The discrete graph itself is π-free; the projection that integrates is where π appears. Existing GeoVac observables (Pauli counts, magic numbers, hydrogenic eigenvalues, Bargmann–Segal lattice, graph-native QED) are confirmatory; the canonical π-bearing observables (vacuum-polarization $\Pi = 1/(48\pi^2)$ and β-function $2\alpha^2/(3\pi)$, the standard flat-space values that Paper 28 enters, and Stefan-Boltzmann) are all evaluated through continuous integrations of the temporal kind.

Beyond the verification sprints KG-1, KG-2, KG-3 and the closed forms reported in Sections 5 and 7.3 (among them the S^5 conformal $-31/60480$ and the Bargmann–Segal harmonic-oscillator $-17/3840$, both computed here), this paper introduces no new computation. Its contribution is the structural observation that *time, not mass*, is the projection axis along which π enters the framework, and that the observer's finite-time measurement window is the formal mechanism.

1 Introduction

GeoVac Paper 34 [12] introduced a two-layer reading of the framework: Layer 1 is a bare combinatorial graph (quantum-number labels, rational matrix entries, algebraic eigenvalues, no physics), and Layer 2 is a finite family of named projections from the bare graph to physical settings. Each projection introduces specific physical variables, specific dimensions, and a specific transcendental signature. Paper 18 [4] provides the parallel transcendental taxonomy: intrinsic, calibration, embedding, algebraic-implicit, composition, inner-factor input. Paper 24 [7] establishes the Bargmann–Segal lattice as bit-exactly π-free; Paper 28 [8] establishes graph-native QED as algebraic in $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$.

The natural question, which Paper 34 leaves open and Paper 31 [9] sharpens through the universal/Coulomb partition, is: where exactly does π enter the framework, and which projections in the Layer 2 dictionary are responsible? Most of the canonical π-bearing observables in GeoVac to date (vacuum polarization, β-functions, Stefan–Boltzmann constants, fine-structure calibration $1/(4\pi)$) have been pinned to specific projections empirically – the Hopf bundle for the α Observation, the Connes–Chamseddine spectral action for one-loop QED, the vector-photon promotion for the $1/(4\pi)$ per-loop Weyl measure on the Hopf base. But there is no single statement of the form "π enters at step X, and at no other step."

This paper makes such a statement, in the context of the simplest non-trivial relativistic system the framework can describe: a free scalar field on $S^3 \times \mathbb{R}$, governed by the Klein-Gordon equation. The Klein-Gordon spectrum is well-known and standard; the contribution here is that, when computed within the GeoVac algebraic-ring tracking discipline, it provides a clean and mechanically verifiable "before / after" for the appearance of π.

The result has a direct consequence for Paper 34's projection dictionary: the row that has been implicitly conflating "introduces a mass" with "introduces an observation window" is underresolved. These are categorically distinct projections on the transcendental axis. The first is ring-preserving; the second injects 2π.

Sprint provenance

The verification computations (KG-1, KG-2, KG-3)[1] are documented in `debug/kg_sprint_memo.md` and the three companion JSON data files; the scripts that produced them are `debug/kg1_algebraic_ring.py`, `debug/kg2_temporal_compactification.py`, and `debug/kg3_casimir_s3_s1.py`. All three completed without blocked items; one minor implementation bug (`sp.nsimplify` mis-rewriting rationals) was identified and fixed during KG-1 and is documented in the memo.

2 Klein-Gordon on $S^3 \times \mathbb{R}$: setup

Take the metric $ds^2 = -dt^2 + R^2 d\Omega_3^2$ where $d\Omega_3^2$ is the round metric on the unit S^3 and R is the spatial radius. The Klein-Gordon equation

$$(\Box + m^2)\phi = 0 \tag{1}$$

separates as $\phi(t, \Omega) = e^{-i\omega t} Y_n(\Omega)$, with Y_n the hyperspherical harmonics.

The scalar Laplace–Beltrami on the unit S^3 has eigenvalues

$$-\Delta_{S^3} Y_n = n(n+2)\, Y_n, \quad n = 0, 1, 2, \ldots, \tag{2}$$

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

with degeneracies $(n+1)^2$. This is standard SO(4) representation theory; the eigenvalues are integers, the multiplicities are perfect squares, and no transcendental content appears. Note that this is the bare scalar Laplacian; it is distinct from the Fock-projected Coulomb operator with eigenvalues $\lambda_n = -(n^2-1)$ used elsewhere in the framework [2].

The dispersion relation from (1) and (2) is

$$\omega_n^2 = \frac{n(n+2)}{R^2} + m^2. \tag{3}$$

Take $R = 1$ throughout (units restored only at the level of physical observables); then $\omega_n = \sqrt{n(n+2)+m^2}$.

3 The algebraic ring (KG-1)

Observation 1 (KG-1: π-free certificate for the bare KG spectrum)**.** *For every $n \in [1, 50]$ and every $m^2 \in \{0, 1, 1/4, 2\}$, the Klein-Gordon eigenvalue $\omega_n = \sqrt{n(n+2)+m^2}$ admits a unique decomposition*

$$\omega_n = c_n \cdot \sqrt{d_n}, \quad c_n \in \mathbb{Q}, \quad d_n \in \mathbb{Z}_{>0} \text{ square-free.} \tag{4}$$

Symbolic verification (`sp.simplify`$(\omega_n^2 - c_n^2 d_n) = 0$*) passes for all 200 cases. No transcendentals (π, e, log, ζ-values) appear in any ω_n for any rational m^2 in the panel.*

The square-free generators that appear vary by mass. A representative slice (first six values of n):

m^2	ω_1	ω_2	ω_3	ω_4
0	$\sqrt{3}$	$2\sqrt{2}$	$\sqrt{15}$	$2\sqrt{6}$
1	2	3	4	5
1/4	$\sqrt{13}/2$	$\sqrt{33}/2$	$\sqrt{61}/2$	$\sqrt{97}/2$
2	$\sqrt{5}$	$\sqrt{10}$	$\sqrt{17}$	$\sqrt{26}$

The $m^2 = 1$ case is structurally special: $n(n+2)+1 = (n+1)^2$, so the spectrum collapses to the integers $\omega_n = n+1$. This is exactly the conformally coupled massless scalar on the unit S^3 — the conformal mass shift ξR at $\xi = 1/6$ on a manifold of constant scalar curvature $R = 6$ (the unit S^3) gives an effective $m^2_{\text{eff}} = 1$ [14]. The collapse is what makes the spectral zeta of the conformally coupled scalar reduce to the ordinary Riemann zeta (Section 5).

For $m^2 \in \{0, 1/4, 2\}$, the union of square-free generators that appear in $n \in [1, 50]$ contains 136 distinct integers ranging from 2 to 10401, all in $\mathbb{Q}[\sqrt{d}]$ for some d. (Recounted 2026-08-28; the previously quoted "134… from 1" was wrong in both the count and the lower endpoint — 1 is not a generator in this set.)

Negative controls. For $m^2 = 1/\pi$ and $m^2 = \sqrt{2}$ (both irrational), the ring is broken at $n = 1$ and the corresponding irrational propagates into ω_n^2 for all n. The mechanism is trivial – the spectrum is a literal sum – but the observation pins the boundary: the Klein-Gordon eigenvalue is in $\mathbb{Q}[\sqrt{d_1}, \sqrt{d_2}, \ldots]$ if and only if m^2 is in the same ring. The mass parameter is, by itself, transcendental-free *when supplied as a rational number*; the framework does not introduce additional π during the parameter projection.

Remark 1 (Comparison with Paper 24 and Paper 28)**.** *Paper 24's π-free certificate for the Bargmann–Segal S^5 HO graph and Paper 28's algebraic-extension certificate for graph-native QED were both established at the level of matrix elements of the discrete graph operators. Observation 1 extends the same kind of certificate to the relativistic spectrum on $S^3 \times \mathbb{R}$, treating the spectrum itself (not just the graph operator) as the object certified. The three certificates partition cleanly: Paper 24 covers the bare HO graph adjacency, Paper 28 covers the finite-graph QED matrix elements, and the present observation covers the relativistic spectrum of a free field on $S^3 \times \mathbb{R}$. None of them require π.*

4 Where 2π enters (KG-2)

The Klein-Gordon equation on $S^3 \times \mathbb{R}$ has the π-free spectrum of Section 3. The Klein-Gordon equation on $S^3 \times S^1_\beta$, where the temporal direction is periodically identified with period β, has the spectrum

$$\omega^2_{n,k} = n(n+2) + \left(\frac{2\pi k}{\beta}\right)^2 + m^2, \quad n = 0, 1, 2, \ldots, \quad k \in \mathbb{Z}. \tag{5}$$

The temporal eigenvalues $\omega^t_k = 2\pi k/\beta$ are the standard Matsubara modes; their factor of 2π is the ratio of one full circle (2π in radian measure) to the circumference β of the identified temporal direction. This factor plays the same structural role as the $1/(4\pi)$ unit-S^2 Weyl (two-dimensional heat-kernel) normalisation of [11]: each is an M1 ($k = 0$ volume slot) volume factor, the calibration constant attached to a compact-direction integral measure.

The before/after is mechanical:

Step	Eigenvalue object	Transcendentals
Spatial Laplace–Beltrami on S^3	$-\Delta Y_n = n(n+2)Y_n$	none (integers)
KG dispersion on $S^3 \times \mathbb{R}$, t open	$\omega^2_n = n(n+2) + m^2$	none (rational m^2)
Periodic time, $k = 0$ sector	$\omega^2_{n,0} = n(n+2) + m^2$	none
Periodic time, $k \neq 0$ sector	$\omega^2_{n,k} = n(n+2) + (2\pi k/\beta)^2 + m^2$	$\boldsymbol{2\pi}$ *appears here*

Observation 2 (KG-2: temporal compactification injects 2π)**.** *The first eigenvalue of* (5) *that contains π is the $(n{=}0, k{=}1)$ Matsubara mode at $\omega^2 = 4\pi^2/\beta^2$. At $\beta = 1$, $m = 0$, the first seven eigenvalues sorted by ascending ω^2 are (listing ω):* 0, $\sqrt{3}$, $2\sqrt{2}$, $\sqrt{15}$, $2\sqrt{6}$, $\sqrt{35}$, 2π. *The first six are in $\mathbb{Q}[\sqrt{d}]$; the seventh is the first π-bearing one. (Corrected 2026-08-28: an earlier version listed five and placed 2π fifth, omitting the $n = 4, 5$ spatial modes $2\sqrt{6} \approx 4.899$ and $\sqrt{35} \approx 5.916$, both below $2\pi \approx 6.283$. The load-bearing claim — that 2π is the* first *π-bearing eigenvalue — is unaffected.) No earlier step in the construction injects π.*

The "injection" is structurally the same statement as the Peter–Weyl realization on S^1: the compact circle of length β has Laplacian eigenvalues $(2\pi k/\beta)^2$ because the smallest non-trivial wavenumber on a circle of length β is $2\pi/\beta$. There is no other source. All higher π^k content in the finite-temperature partition function, free energy, and Casimir energy on $S^3 \times S^1_\beta$ traces back to multiple Matsubara modes multiplying.

5 Casimir energy on S^3: a one-projection match (KG-3)

The Casimir energy of a conformally coupled massless scalar on the unit S^3 (no temporal compactification: temperature $T = 0$) is zeta-regularized as

$$E_{\rm Cas} = \frac{1}{2}\zeta_X(-1), \quad \zeta_X(s) = \sum_{n=0}^{\infty}(n+1)^2\,\omega_n^{-s}, \tag{6}$$

where the spectrum and degeneracy are those of the conformally coupled massless scalar. By the integer collapse of Section 3, $\omega_n = n + 1$ and the spectral zeta reduces to the Riemann zeta shifted by two arguments:

$$\zeta_X(s) = \sum_{n=0}^{\infty}(n+1)^{2-s} = \zeta_R(s-2). \tag{7}$$

Evaluating at $s = -1$:

$$E_{\rm Cas} = \frac{1}{2}\zeta_R(-3) = \frac{1}{2}\cdot\frac{1}{120} = \boxed{\frac{1}{240}} \tag{8}$$

where $\zeta_R(-3) = -B_4/4 = 1/120$ via the Bernoulli number identity ($B_4 = -1/30$). The result is exact rational, with no transcendental content at any step.

Observation 3 (KG-3: spatial Casimir on S^3 is π-free)**.** *The ζ-regularized Casimir energy of the conformally coupled massless scalar field on the unit S^3 is the exact rational $E_{Cas} = 1/240$. The full per-step transcendental ledger is:*

Step	*Quantity*	*Transcendentals*
Spatial spectrum (conformal)	$\omega_n = n+1$, *deg.* $(n+1)^2$	*none*
Spectral zeta	$\zeta_X(s) = \zeta_R(s-2)$	*none*
Evaluate $\zeta_R(-3)$	$-B_4/4 = 1/120$	*none*
Casimir	$E_{Cas} = 1/240$	*none*

The match against the textbook value (Bytsenko et al. [15], Dowker–Critchley [16], Ford [17]) is exact rational equality, verified to 60 decimal places numerically.

The structural significance of Observation 3 is that the spatial Casimir on S^3 provides Paper 34 with a clean, exactly computable, single-projection match: the projection invoked is the Fock conformal projection (Paper 34 Table 1 row 1) plus the conformal coupling shift; the transcendental signature is rational; and the value 1/240 matches an independently established physics quantity. By the Layer-2-presence bound of Paper 34 (this is a zero-Layer-2-input, rational-analytic chain), a single-projection match of this kind should sit at the framework's basis-quality limit – here exact – and indeed it is.

Observation 4 (The dimensional Casimir ladder: S^1, S^3, S^5 and the Bernoulli rungs (Sprint GB-3))**.** *The conformal-scalar Casimir on S^{2k-1}, $E_{Cas} = \frac{1}{2}\sum_l \omega_l\,\deg_l$ with $\omega_l = l + (d-1)/2$, leads with $\zeta_R(-(2k-1))$, i.e. the Bernoulli number B_{2k}. The first three rungs:*

Manifold	$\omega\cdot\deg$ *(in n)*	E_{Cas}		*rung*
S^1	n	$\frac{1}{2}\zeta(-1)$ $-1/24$	=	B_2
S^3	n^3	$\frac{1}{2}\zeta(-3)$ $1/240$	=	B_4
S^5	$(n^5-n^3)/12$	$\frac{1}{24}(\zeta(-5)-\zeta(-3))$ $-31/60480$	=	B_6 (+ B_4)

On S^3 the degeneracy n^2 keeps $\omega \cdot \deg$ a single power (n^3), so the Casimir is a clean single rung. On S^5 the quartic degeneracy $n^2(n^2-1)/12$ spreads $\omega \cdot \deg$ across n^5 and n^3, so the S^5 Casimir carries both B_6 *($\zeta(-5) = -1/252$, coefficient $+1/24$) and B_4 ($-1/24$), with B_2* absent. *The value $-31/60480$ is exact (verified in* `sympy`*) and, to our knowledge, newly computed; S^5 is a GeoVac manifold via the Bargmann–Segal lattice (Paper 24). The spreading is the same multiplicity mechanism that obstructs the Dirac-zeta functional equation (Paper 28's functional-equation obstruction), and the S^3/S^5 rungs place the framework's Casimirs on the Bernoulli ladder of Paper 32 (the two-layer split read as the ζ functional-equation reflection $\zeta(s) \leftrightarrow \zeta(1-s)$, with the transcendental window $\zeta(2k) = \text{rational} \cdot \pi^{2k}$ supplying the M2 observables and the negative-integer window $\zeta(1-2k)$ the π-free skeleton). Source:* `debug/sprint_gb3_b6_rung_memo.md`.

5.1 The Stefan–Boltzmann constant lives in the temporal projection

Compactifying the temporal direction to S^1_β promotes the Casimir computation to the finite-temperature free energy. The high-temperature expansion contains the Stefan–Boltzmann term

$$\frac{F(\beta)}{V_3} \sim -\frac{\pi^2/90}{\beta^4} + \cdots, \qquad \beta \to 0, \tag{9}$$

where $V_3 = 2\pi^2$ is the volume of the unit S^3. The $\pi^2/90$ traces directly to the Matsubara sum: after analytic continuation, the temporal integral collapses to $\zeta_R(4) = \pi^4/90$, which is exactly the calibration tier of Paper 18 (even-zeta content from a second-order operator on a compact direction). The Matsubara prefactor $(2\pi/\beta)^{2s}$ from (5) provides one factor of π^{2s}, and the $\zeta_R(4)$ provides the rest; the net π-power survives to give the Stefan–Boltzmann constant.

Remark 2 (The high-T $\pi^2/90$ is Matsubara-induced)**.** *The $\pi^2/90$ in the Stefan–Boltzmann limit of the $S^3 \times S^1_\beta$ free energy is not a property of the spatial spectrum on S^3. It is the result of integrating the spatial spectrum against a Matsubara temporal sum, where the latter's 2π injection (Observation 2) propagates through the $\zeta_R(4)$-controlled spectral integral. Without the temporal compactification, the same spatial spectrum gives Casimir $1/240$ (Observation 3), with no π.*

Observation 5 (Two-window discriminator: each Bernoulli rung is lit from both sides by distinct physics (Sprint GB-4))**.** *Remark 2 is the B_4 instance of a general structure. The conformal-scalar Casimir on S^{2k-1} supplies the* skeleton *window $\zeta_R(1-2k)$ (vacuum, π-free); the Stefan–Boltzmann free energy on $S^{2k-1} \times S^1_\beta$ supplies the* transcendental *(M_2) window $\zeta_R(2k) = \text{rational} \cdot \pi^{2k}$ (thermal). The Bose–Einstein integral $\int_0^\infty x^d/(e^x-1)\,dx = \Gamma(d+1)\zeta_R(d+1)$ makes this explicit: $d = 3$ gives $\zeta_R(4) = \pi^4/90$ (B_4), $d = 5$ gives $\zeta_R(6) = \pi^6/945$ (B_6).*

rung	*skeleton (vacuum)* $\zeta(1-2k)$	M_2 *$\zeta(2k)$ (thermal)*
B_4	$\zeta(-3) = 1/120$, S^3 *Casimir*	$\zeta(4) = \pi^4/90$, $S^3 \times S^1$ *S–B*
B_6	$\zeta(-5) = -1/252$, S^5 *Casimir*	$\zeta(6) = \pi^6/945$, $S^5 \times S^1$ *S–B*

Each rung is thus lit from both *functional-equation windows by* physically distinct *observables — vacuum Casimir and thermal radiation — related by the temperature-inversion duality $T \leftrightarrow 1/T$, which is the ζ functional equation $\zeta(s) \leftrightarrow \zeta(1-s)$ on the thermal circle.*

Negative control (the discriminating half). *The genuinely combinatorial GeoVac invariants — those set by the label data rather than by sphere field theory — are* off *the ladder: $B = 42$, $\Delta = 1/40$*

(Paper 2), and the matched scale $\kappa = -1/16$ (matched to the graph band edge, not forced by the packing axiom) are not ζ-rung values (and the correlation entropy $S_{\text{full}}(\text{GS})$ is PSLQ-null off the master Mellin engine, Sprint TD Track 5). So the ladder is the functional-equation structure of the framework's spectral-geometry *sector (Casimir + thermal + heat-kernel),* not *a universal property of every GeoVac quantity.*

Honest scope. *Both windows are lit by sphere field theory, so the positive landings are textbook in isolation (Planck blackbody, Casimir, Cardy temperature inversion); the GeoVac content is (i) that the framework's gravity and thermal sectors realize this two-window structure at three rungs, and (ii) the clean boundary against the combinatorial core. This sharpens, but does not exceed, the structural correspondence of Paper 32's Bernoulli-ladder remark; the non-trivial zeros remain untouched (Paper 28's functional-equation obstruction). As a cross-connection, the on-ladder/off-ladder split ($F = \zeta(2)$ on, B and Δ off) is exactly the Phase-4G "no common generator" decomposition of $K = \pi(B + F - \Delta)$ viewed from the ladder. Source:* `debug/sprint_gb4_discriminator_memo.md`.

6 Implications for Paper 34

Paper 34's projection dictionary, in the form current when this sprint ran, did not separately index "introduce a mass parameter into the spectrum" from "compactify the temporal direction for finite-time observation." In practice these had been bundled with Fock-conformal (which carries Z and E) or with the spectral action (which carries Λ). This sprint shows the bundling is too coarse: the mass projection is ring-preserving, while the observation/temporal-window projection injects 2π *at the spectrum level, not just at the integration level.* These are structurally different and should be split.

Proposition 1 (Two new rows for the Paper 34 projection dictionary)**.** *The Paper 34 projection list should include the following two explicit rows:*

- ***Rest-mass projection*** *(§ XIV.A). Source: bare KG spectrum on $S^3 \times \mathbb{R}$. Target: massive KG spectrum $\omega_n^2 = n(n+2)/R^2 + m^2$. Variables introduced: m (mass; equivalently rest-energy mc^2 at $c = 1$). Dimension introduced: mass = energy at $c = 1$. Transcendental signature:* trivial*; the bare-graph algebraic-extension ring is preserved when $m^2 \in \mathbb{Q}$ (or any ring extension of $\mathbb{Q}$ already present in the spatial spectrum). Used in: muonic / electronic / positronium reduced-mass corrections; nuclear-electronic composed Hamiltonian (Paper 23 hyperfine I.S coupling); any future Klein-Gordon or Dirac-on-S^3 relativistic mass calculation.*
- ***Observation / temporal-window projection*** *(§ XIV.B). Source: KG spectrum on $S^3 \times \mathbb{R}$. Target: KG spectrum on $S^3 \times S^1_\beta$ with periodic Euclidean time of period β (equivalently, a finite-time observation window of duration β in real time). Variables introduced: β (or temperature $T = 1/\beta$). Dimension introduced: time (= inverse energy). Transcendental signature: $2\pi \cdot \mathbb{Q}$ per Matsubara mode in the spectrum directly; in spectral integrals over the compactified direction, $\pi^{2k} \cdot \mathbb{Q}$ from $\zeta_R(2k)$ values, including the Stefan-Boltzmann $\pi^2/90$. Used in: all finite-temperature observables (Casimir at $T \neq 0$, free-energy density, Stefan-Boltzmann constants); all QED loop integrals that involve continuous time integration; any spectral-action evaluation of $\mathrm{Tr}\, f(D^2/\Lambda^2)$ where the trace is over a compact-time direction.*

The two rows together replace the implicit "mass / observation" content that has been bundled with other rows. They make explicit a distinction that is structural in the framework but had not been named. The Casimir computation in Section 5 is the cleanest empirical demonstration: changing the temporal boundary condition (open $\mathbb{R} \to$ closed S^1_β) changes the transcendental class

of the computed observable from rational $(1/240)$ to $\pi^{2k}\cdot\mathbb{Q}$ (Stefan-Boltzmann), without any change to the spatial spectrum.

Remark 3 (This is a no-edits-yet proposal)**.** *Per the Paper 34 maintenance protocol, the addition of two new rows to the projection dictionary is a substantive structural change and requires PI direction. The PM may append match rows to Paper 34's empirical catalogue autonomously, but new projection-list rows are not PM-autonomous. The present paper records the structural finding and the proposal; Paper 34 itself is not amended.*

7 Falsifiable prediction

Prediction 1 (π enters at temporal-projection compactifications, only)**.** *A GeoVac observable contains π if and only if its evaluation includes a continuous integration over a temporal or spectral parameter that has been promoted from the discrete graph spectrum. Equivalently: the discrete graph itself is π-free; the projection that* integrates *(over time, over a Matsubara mode, over a Mellin parameter, over an analytically continued spectral mode) is where π appears.*

The closest theorem-grade analog in the continuum literature is the **Bisognano–Wichmann theorem** [21] (see [19] for a contemporary geometric account in the Morinelli–Neeb–Ólafsson lineage and [20] for the non-unitary CFT extension): the BW theorem identifies the modular automorphism of a wedge-restricted field algebra with the Lorentz-boost flow at rapidity 2π per unit modular parameter, which is exactly the compactification operation Prediction 1 flags as the unique π-source. Section 8.6 below develops this correspondence (an analogy) explicitly, and Paper 32 § VIII makes the conditional spectral-triple-side case-exhaustion statement that anchors the prediction's integrations in three mechanisms (it does not derive the prediction). The reader who treats Prediction 1 as a discrete-substrate correspondence (analogy) with BW will recover the falsifier (§ 8.6 Track D #1) and the 208/208 verification panel below from a single conceptual root.

7.1 Confirmatory evidence

Six existing GeoVac observables that do not require integration over a continuous parameter are π-free, as predicted (the sixth was added after sprint KG-5):

1. **Pauli term counts** (Paper 14): the $N_{\text{Pauli}} = 27.90 \cdot Q$ rule (main-group; $30.03 \cdot Q$ d-block; exact global-M_L rule) across 35 composed molecules; the counts are integers at every Q.
2. **Nuclear magic numbers** (Paper 23): $\{2, 8, 20, 28, 50, 82, 126\}$ from HO + spin-orbit graph eigenvalue counting; all integers.
3. **Hydrogenic atomic eigenvalues via the Fock/S^3 projection** (Paper 7): $\lambda_n = -(n^2-1)$ for the Fock-projected Coulomb operator (a continuum S^3 Laplace–Beltrami property, not a bare-graph one); pure integers at every n.
4. **Bargmann–Segal HO graph** (Paper 24): bit-exactly π-free in exact rational arithmetic at every finite $N_{\max}$, verified for $N_{\max} \leq 5$.
5. **Graph-native QED matrix elements** (Papers 28, 33): every matrix element in $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$ at every finite $n_{\max}$.
6. **Spatial Dirac Casimir on unit** S^3 (Section 9 item 1): $E_{\text{Cas}}^{\text{Dirac}} = 17/480$, exact rational, computed in sprint KG-5 against the Camporesi–Higuchi spectrum. Confirms the prediction in the spinor sector.

7.2 Counter-examples (also confirmatory)

Three existing GeoVac observables that *do* contain π, all of which arise from a temporal or spectral continuous integration:

1. **Vacuum polarization** $\Pi = 1/(48\pi^2)$ (Paper 28): standard one-loop QED (the flat-space value, entered in Paper 28, not computed from S^3 data), evaluated as a continuous integral over the Schwinger proper time $t \in (0, \infty)$. The proper-time integration is the temporal-projection step; the π^2 is the four-dimensional heat-kernel normalisation $(4\pi)^{-2}$. (An earlier version traced it to an S^3 coefficient $a_2 = \sqrt{\pi}/8$, squared; on S^3 the Dirac a_2 vanishes and that value was a sign error, corrected 2026-10-02, Paper 28.)
2. **QED β-function** $2\alpha^2/(3\pi)$ (Paper 28): same provenance: the π in the denominator comes from the same four-dimensional loop normalisation.
3. **Stefan-Boltzmann** $\pi^2/90$ (Section 5): high-T limit of the $S^3 \times S^1_\beta$ free energy density; π from the Matsubara $\zeta_R(4)$ sum.

In every case, the π-bearing observable can be traced to a specific continuous integration, and the integration can be identified with the observation/temporal-window projection of Proposition 1. The framework so far has no exception to the prediction.

7.3 Sprint TX-B verification panel (May 2026)

Sprint TX-B (May 2026) tested Prediction 1 on five new observables, with predictions logged a priori in `debug/data/tx_b_predictions.json` *before any computation script was run.* This discipline guards against retroactive reframing, the standard failure mode for falsifiable predictions (cf. `docs/curve_fit_audit_memo.md`, 2026-05-02). The panel was designed to probe four corners of the prediction:

1. **Stefan-Boltzmann coefficient on** $S^3 \times S^1_\beta$ (positive control, π predicted): re-derives $\pi^2/90$ from the Matsubara sum. Confirmed.
2. **Spatial Casimir on Bargmann–Segal** S^5 **for the 3D HO** (positive prediction, no π): the Bargmann construction is a different geometry from the S^3 Coulomb / S^3 Dirac sectors of KG-3 / KG-5. If the half-integer Hurwitz mechanism is structural, the S^5 HO Casimir should produce another exact rational. Computed via the spectral zeta

$$\zeta_X(s) = \tfrac{1}{2}\big[\zeta_R(s{-}2,\, 3/2) - \tfrac{1}{4}\zeta_R(s,\, 3/2)\big]$$

at $s = -1$. Bernoulli-at-half-integer values $B_2(3/2) = 11/12$, $B_4(3/2) = 127/240$ give Hurwitz at negative integers $\zeta_R(-1, 3/2) = -11/24$, $\zeta_R(-3, 3/2) = -127/960$, and the bracket $[\zeta_R(-3, 3/2) - \tfrac{1}{4}\zeta_R(-1, 3/2)] = -17/960$. Hence

$$E^{\mathrm{HO},S^5}_{\mathrm{Cas}} \;=\; \tfrac{1}{2}\,\zeta_X(-1) \;=\; -\tfrac{17}{3840} \quad \text{(exact rational, no } \pi\text{)}.$$

This is a clean new closed form, structurally a sibling of KG-3 (scalar S^3 Casimir $= 1/240$) and KG-5 (Dirac S^3 Casimir $= 17/480$). The shared mechanism across all three sectors — half-integer Hurwitz at shift $a = 3/2$, Bernoulli polynomials at $3/2$ rational — is the universal structural driver: any spectrum of the form $E_n = n + a$ with $a \in \mathbb{Q}$ and degeneracy polynomial in n produces a rational Casimir through ζ-regularization, regardless of whether the underlying manifold is S^3 scalar, S^3 Dirac, or S^5 HO. The Coulomb/HO/Dirac asymmetry that Paper 24 highlights at the level of the Laplacian role *does not* extend to the Casimir transcendental class. Confirmed.

3. **Hydrogen Bohr spectrum** (negative control): pure algebraic formula, no integration, no π. Trivial but essential. Confirmed.
4. **Heisenberg-Euler 1-loop coefficient** (positive prediction): the textbook leading coefficient $\alpha^2/(45\pi m_e^4)$ contains π in the denominator, sourced by the Schwinger proper-time integration $\int_0^\infty ds/s^3\, e^{-m^2 s} f(eF, s)$ — the canonical example of an observation/temporal-window projection. We did not derive the absolute coefficient natively from the GeoVac framework (that would require LS-7-style two-loop SE machinery), so this entry is flagged as structural identification (error code S in Paper 34's classification): the transcendental class matches, the absolute value is from the textbook. Confirmed at structural level.
5. **Static Coulomb-like potential on the Hopf S^3 graph** (positive prediction, no π): the regularized discrete Green's function $(L+I)^{-1}$ at finite $n_{\max}$ is built by exact matrix inversion on a finite integer-valued graph. No continuous integration appears. Computed in exact sympy rational arithmetic at $n_{\max} = 2, 3$: every entry is a sympy `Rational`, no π anywhere. The continuum limit ($n_{\max} \to \infty$) would inject π via spherical-harmonic normalization on S^3, exactly where Prediction 1 predicts. Confirmed at finite truncation.

Tally: 5 of 5 confirmed. Combined with the prior KG sprint (200 KG spectrum cases, KG-2 temporal injection, KG-3 scalar Casimir, KG-5 Dirac Casimir), Prediction 1 has now been tested on a cumulative **208 individual checks across the KG and TX-B sprints, with 100% match rate**.

Remark 4 (graduation to load-bearing principle)**.** *With the TX-B verification panel in hand, Prediction 1 graduates from a falsifiable* conjecture *of the framework to a* load-bearing principle *of it. The graduation is two-fold: (i) the prediction has survived 208 independent checks across two sprints, with no exceptions; (ii) it now plays an operational role alongside Paper 18's transcendental cataloguing as a discipline for evaluating new derivations. If a step in a derivation is supposed to produce π, the discipline is to identify the continuous integration over a temporal or spectral parameter that produces it; if no such integration is identifiable, the π is either an error or an unidentified projection (in the latter case requiring an addition to Paper 34's projection list).*

7.4 Falsification criterion

A counter-example to Prediction 1 would be a GeoVac observable whose evaluation does not include a continuous integration over a temporal or spectral parameter, but which contains π. The most natural place to look is the Paper 2 Observation $K = \pi(B + F - \Delta) \approx 1/\alpha$, in which the π is explicit on the right-hand side and the projections involved (finite Casimir trace at $m = 3$, infinite scalar Dirichlet at $s = d_{\max}$, single-level Dirac mode degeneracy) all appear, on present analysis, to be discrete-graph quantities. If the projection-depth analysis of Paper 34 § 9 item 3 is correct, then either (a) the π in K is not what it looks like (i.e., it factors out cleanly as a temporal-integration prefactor), or (b) the present prediction is false and there is a non-temporal route to π in the framework that we have not identified. Either resolution would be informative.

7.5 Anchoring in the spectral triple (conditional; Sprint TS, May 2026)

Track TS-D of Sprint TS (May 2026; memo `debug/track_ts_d_paper35_triple_test_memo.md`) tested Prediction 1 against each of Paper 34's fifteen then-catalogued projections through the spectral-triple lens of Paper 32 [10], and found that every π-source identifiable in any projection reduces to one of three triple-framing mechanisms: M1 ($k = 0$ volume slot, $\mathrm{Vol}(S^2)/4$), M2 (Seeley–DeWitt heat-kernel coefficient $\sqrt{\pi}$ on S^3), or M3 (vertex-topology Hurwitz Dirichlet L-content).

Track TS-E1 sharpened the case exhaustion to a formal theorem statement, now stated in Paper 32 § VIII as the *π-source case-exhaustion theorem* ([**CONDITIONAL**]). Track TS-E3 further showed that M1, M2, and M3 are not three independent mechanisms but three sub-cases of a single master Mellin mechanism $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$ parametrized by operator order $k \in \{0, 1, 2\}$ — the master-mechanism reading is recorded in Paper 18 § III.7.

The case-exhaustion theorem argues, case by case over the Paper 34 projections § III.1–§ III.15 (the fifteen then catalogued), that "the chain contains a continuous spectral/temporal integration" implies "the chain engages one of M1/M2/M3"; the link to "the converged value contains π" is Prediction 1 itself (empirical), and the converse direction is withdrawn as not established (engaging a mechanism is not known to force π). *Within that scope* the theorem anchors the integrations the prediction refers to in the GeoVac spectral triple's three operating modes (Hopf decomposition of the spinor bundle, heat kernel of D^2, vertex-parity selection on the Camporesi–Higuchi spectrum); it does not by itself turn the prediction into a structural corollary. For observables outside the theorem's enumerated scope (the projections added since, and the general biconditional over *all* GeoVac observables) the reading remains the *structural Observation* of this paper — WH7 is registered, not proven — with the 208/208 confirmation as its empirical foundation; where the theorem applies it anchors the integrations in three named mechanisms, without deriving the prediction.

8 Tensor-product verification on $T_{S^3} \otimes T_{S^1_\beta}$

Sprint TD Track 1 (May 2026) constructs the explicit tensor-product spectral triple $T_{S^3} \otimes T_{S^1_\beta}$ with Dirac operator

$$D_{\text{total}} \;=\; D_{S^3} \otimes 1 \;+\; \gamma_{S^3} \otimes D_\tau, \tag{10}$$

where D_τ on S^1_β is the Matsubara operator with bosonic spectrum $\omega_k = 2\pi k/\beta$ or fermionic spectrum $\omega_k = 2\pi(k + 1/2)/\beta$. The anti-commutator $\{D_{S^3} \otimes 1,\, \gamma_{S^3} \otimes D_\tau\} = 0$ makes D^2_{total} block-diagonal (no cross term), so spectra add and the heat kernel factorises as $\mathrm{Tr}\, e^{-sD^2_{\text{total}}} = K_{S^3}(s) \cdot K_{S^1_\beta}(s)$.

8.1 Stefan–Boltzmann as exact $M_1 \times M_2$

The textbook free-energy density for a single bosonic massless scalar on flat $\mathbb{R}^3 \times S^1_\beta$ is $F/V = -(\pi^2/90)\, T^4$. The high-T continuum limit on $S^3 \times S^1_\beta$ recovers the same coefficient. Sprint TD Track 1 derives this as a single SYMPY-exact identity:

$$\frac{F}{V} \;=\; -\;\underbrace{\frac{1}{2\pi^2}}_{M_1 \text{ volume slot}} \cdot \frac{1}{3} \cdot \underbrace{6\,\zeta(4)}_{M_2 \text{ Riemann } \zeta} \cdot T^4 \;=\; -\,\frac{\pi^2}{90}\, T^4 \tag{11}$$

with sympy residual 0. The $1/(2\pi^2)$ is the M_1 volume-slot factor $\mathrm{Vol}(S^2)/(2\pi)^3$; the $6\zeta(4) = \pi^4/15$ is the M_2 Riemann $\zeta(4)$ at the Bose–Einstein integrand. The net π^2 in $-\pi^2/90$ is the $M_1 \times M_2$ product in π-power arithmetic ($\pi^{-2} \cdot \pi^4 = \pi^2$). For Dirac fermions, the same factorisation with the fermionic $\eta(4)/\zeta(4) = 7/8$ ratio gives $F/V_{\text{per-Weyl}} = -(7\pi^2/720)\, T^4$, again with residual 0.

8.2 Spatial Casimir is rational on both bundles

In the zero-T limit the thermal partition function reduces to the spatial Casimir on S^3, recovering Paper 35 KG-3 and KG-5 verbatim:

$$E_{\mathrm{Cas}}^{\text{conformal scalar},\,S^3} = \tfrac{1}{240},$$
$$E_{\mathrm{Cas}}^{\text{full Dirac},\,S^3} = \tfrac{17}{480},$$

both *rational*. The half-integer Hurwitz / Bernoulli-polynomial collapse mechanism noted in §V transports verbatim to the tensor product: the spatial factor contributes no π at the discrete-mode level, and π enters only through the Mellin transform on s when the heat kernel is evaluated.

8.3 Modular residual on the tensor product

The Sprint MR-B closed-form modular residual on the spatial Dirac sector,

$$\varepsilon(t) \;=\; \sum_{m\geq 1}(-1)^m\,\sqrt{\pi}\,e^{-m^2\pi^2/t}\left[\,t^{-3/2} - 2m^2\pi^2\,t^{-5/2} - \tfrac{1}{2}\,t^{-1/2}\right], \tag{12}$$

transports verbatim to the tensor product (it lives on K_{S^3}). The temporal factor contributes a leading $K_{S^1_\beta}(s) \approx \beta/(2\sqrt{\pi s})$ from Jacobi-theta inversion. Crucially, the $\sqrt{\pi}$ in $K_{S^1_\beta}$ is the M_1 $\mathrm{Vol}(S^1)$ / *Mellin-measure* factor, whereas the $\sqrt{\pi}$ in $\varepsilon(t)$ is the M_2 *Seeley–DeWitt* prefactor on S^3. Same numerical constant; structurally distinct mechanisms; distinct tensor factors. The ring of the full modular residual is $\sqrt{\pi}\times\bigoplus_k \pi^{2k}\cdot\mathbb{Q}$ (all M_2 once the $\sqrt{\pi}$ heat-trace normalisation is divided out) on the spatial side, multiplied by $\sqrt{\pi}/\beta\cdot\mathbb{Q}$ (M_1) on the temporal side.

8.4 Transcendental audit holds Prediction 1

Sprint TD Track 1 audits ten distinct transcendental sources in the construction. By tensor factor: 4 π-bearing on S^3 (volume-slot factor, $\mathrm{Vol}(S^3)/\pi^2$, M_2 modular exponent π^2, M_2 Seeley–DeWitt prefactor $\sqrt{\pi}$); 5 on S^1_β (Matsubara modes, Bose–Einstein integrand, theta-inversion prefactor, volume-slot factor, fermionic $\eta(4)$ ratio); 3 rational with no π (scalar Casimir 1/240, Dirac Casimir 17/480, fermionic 7/8 ratio). By mechanism: 3 M_1, 4 M_2, 0 M_3 (no vertex coupling in the free thermal partition function), 1 $M_1\times M_2$ product (Stefan–Boltzmann), 2 rational collapses (M_2 at half-integer Hurwitz).

Every π has a continuous-integration source: Matsubara compactification of imaginary time, Weyl-asymptotic spatial integration, Mellin transform of the heat-kernel trace, Jacobi-theta inversion, anti-periodic boundary condition. Every rational result comes from a discrete spectral evaluation: Bernoulli polynomials at $a = 3/2$, Dirichlet $\eta(4)/\zeta(4)$ rational ratio, half-integer Hurwitz collapse. **Prediction 1 holds verbatim on the tensor product.**

8.5 Honest scope

This subsection is a verification of the master Mellin engine partition on a known thermal field-theory example. The Stefan–Boltzmann constant is textbook; the new content is the explicit factor-by-factor structure that pins each π to its tensor factor and to its Mellin mechanism. The construction is at the spectrum level; a full spectral-triple-of-tensor-products theorem in the sense of Paper 39 (which handles the same-manifold case $T^{\lambda_a}_{S^3}\otimes T^{\lambda_b}_{S^3}$) is not claimed for the cross-manifold $S^3\times S^1_\beta$ case. See Paper 32 §VIII.D on the cross-manifold open question. Sprint TD Track 1's `geovac/thermal_tensor_triple.py` (SYMPY symbolic, 36 tests) is the computational anchor.

8.6 Bisognano–Wichmann reading of the temporal-window projection

Sprint TD Track 4 (`debug/sprint_td_track4_memo.md`, May 2026) extends the spectrum-level construction of § 8 to the Euclidean Schwarzschild cigar. Smoothness at the horizon $r = 2M$ (the standard Hawking–Gibbons absence-of-conical-defect argument) forces the imaginary-time circle to have period $\beta_{\text{cigar}} = 8\pi M$; with M supplied by the Einstein equations, this β is posited rather than generated by the framework, and the M1 ($k = 0$ volume slot) / $\mathrm{Vol}(S^1_\tau)$ reading re-expresses the continuum Hawking temperature $T_H = 1/(8\pi M)$ (a consistency reproduction, not a derivation). Sprint TD Track 4's `matsubara_spectrum()` called with $\beta = 8\pi M$ reproduces the standard Hawking spectrum bit-identically (lowest bosonic mode $1/(4M) = \kappa_g$ surface gravity; lowest fermionic πT_H); sympy residual zero on all three identities ($\beta = 8\pi M$, $T_H = \kappa_g/(2\pi)$, $\beta = 4M \cdot 2\pi$).

Track D (`debug/bisognano_wichmann_track_d_memo.md`, May 2026) makes explicit the Bisognano–Wichmann reading of this result. The Bisognano–Wichmann theorem [21] (see also [19] for a contemporary geometric perspective in the Morinelli–Neeb–Ólafsson lineage) states that the modular automorphism of the algebra of observables localized in a Rindler wedge, with respect to the vacuum state, IS the unitary representation of the Lorentz boost subgroup that maps the wedge to itself; the modular Hamiltonian K generates the boost, and the Tomita–Takesaki KMS condition forces the analytic period $\sigma_{i\cdot 2\pi} = \text{identity}$. Sewell 1982 [22] extended this to spacetimes with bifurcate Killing horizons (for spacetimes admitting the requisite analytic continuation; the rigorous general result is Kay–Wald 1991 [23]): the natural Hartle–Hawking vacuum [24] restricted to the Schwarzschild exterior is a KMS state at inverse temperature $\beta = 8\pi M$, and the Tomita modular automorphism is the surface-gravity-rescaled Killing flow. Hartle–Hawking [24] provided the path-integral derivation of the same imaginary-time period $8\pi M$.

The structural correspondence. Under the Wick-rotation chain [Hartle–Hawking 1976] → [Sewell 1982] → [Bisognano–Wichmann 1976], the framework's M1 2π on S^1_τ matches the period of the modular-flow / Lorentz-boost orbit of a stationary observer outside the Schwarzschild horizon. Equivalently, the temporal-window projection (Paper 34 §III.15), instantiated at $\beta = 8\pi M$, admits two readings of one number: a volume factor on the temporal S^1_τ (Riemannian; the $k = 0$ volume slot) and, by analogy, a modular-flow / Lorentz-boost orbit period (Lorentzian, via the standard Wick rotation). This is a *structural correspondence* (an analogy), not a literal identification: two mathematical objects (the framework-side volume factor; the Lorentzian-side modular-automorphism period) are matched by the published Wick-rotation prescription, not by an operator-level proof inside the spectral triple; identifying the framework's generator with a boost is open (Paper 42's O0). The continuum side of the correspondence stands on the published QFT machinery, which is mature and verified independently of GeoVac.

Unruh corollary. The same M1 reading applied to a Rindler wedge is consistent with the Unruh effect: a uniformly accelerated observer at proper acceleration a (external input) has the continuum Wick-rotated $\beta_{\text{Rindler}} = 2\pi/a$, which the M1 volume slot re-expresses as $T_U = a/(2\pi)$ without deriving it. Sprint TD Track 4's `matsubara_spectrum()` machinery applies verbatim with β_{Rindler} in place of β_{cigar}; the pendant check flagged in Track D §5.4 is carried out in § 8.7.

Honest scope. Track D commits to four claims: (1) the continuum $\beta = 8\pi M$ on the cigar, posited by horizon regularity with M supplied by the Einstein equations, is consistent with the framework's M1 ($k = 0$ volume slot) reading (sympy residual zero, Sprint TD Track 4) — a consistency reproduction, not a derivation of β; (2) this β is the imaginary-time period of the Hartle–Hawking thermal state; (3) Sewell 1982 shows (given the requisite analytic continuation; Kay–Wald 1991 in general) that the modular flow of this thermal state IS the Killing flow, the Schwarzschild analogue of the Bisognano–Wichmann boost; (4) the framework-side 2π and the Lorentzian-side modular-flow 2π are the same number, by the published Wick-rotation chain. It

does *not* claim: an operator-level proof that the modular Hamiltonian on the truncated metric spectral triple closes at period 2π (this is the principal falsifier, a follow-up beyond sprint scale); a Lorentzian-signature spectral triple (a major NCG structural extension, Track B verdict NO-GO; `debug/lorentz_boost_scoping_memo.md`); a derivation of the cigar geometry from the GeoVac packing axiom (Sprint TD Track 4 V3 NEGATIVE; the cigar is forced by Einstein equations + Schwarzschild horizon, not by GeoVac); Bekenstein–Hawking entropy from the horizon S^2 (out of scope, Sprint TD Track 4 V2 deferred).

Metric-level dual (Paper 45 § open Q1, 2026-06). The temporal-compactification picture here has a metric-level counterpart. The posited wedge generator of Paper 42 (its identification with the Bisognano–Wichmann boost is open, Paper 42 O0) is *compact* (integer spectrum, $e^{2\pi i K} = I$ at every finite cutoff, the same period-2π closure) [SYMBOLIC]. The further reading has two legs (Paper 45 open Q1). That the finite-cutoff quantum metric on the truncated Krein triple is therefore Euclidean (forward-triangle), with the genuine reverse-triangle signature confined to the non-compact continuum limit, is [ARGUED] and generator-independent. That the Lorentzian signature is the Wick rotation of the compact KMS 2π-circle, a convention, is [ARGUED] and, for the Bisognano–Wichmann generator, conditional on O0. On that reading, just as compactification is the one step that injects π into the *spectrum* (this paper), it is the one step that fixes the *signature* as a Wick-rotation convention: the same compactness $\Rightarrow$ discreteness $\Rightarrow$ π/β reading on both faces.

Cross-references. Paper 32 §VIII Remark `rem:bisognano_wichmann_reading` for the spectral-triple-side statement and case-exhaustion-theorem context; Paper 34 §III.15 footnote for the temporal-window projection's Bisognano–Wichmann reading; Paper 34 §V.B for the Hawking-T off-precision catalogue row (error class C, M external Einstein-equations input); Paper 45 § open Q1 for the metric-level dual above.

8.7 Unruh pendant: the third Wick-rotation face

The same M1 reading applied to a Rindler wedge is consistent with the Unruh effect (it re-expresses, and does not derive, the continuum β_U), the flat-space analog of Hawking radiation and the third face of the Wick-rotation chain that § 8.6 traversed in the Schwarzschild geometry. Sprint Unruh-Pendant (`debug/unruh_pendant_memo.md`, May 2026, a sprint-scale pendant follow-up to Track D) documents the construction.

Wick rotation of Rindler. A uniformly accelerated observer at proper acceleration a traces a hyperbolic worldline in the right Rindler wedge $x > |t|$ of Minkowski space; in Rindler coordinates (η, ρ) defined by $t = \rho \sinh(a\eta)$, $x = \rho \cosh(a\eta)$, the wedge metric is $\mathrm{d}s^2 = -\rho^2\,\mathrm{d}\eta^2 + \mathrm{d}\rho^2 + \mathrm{d}y^2 + \mathrm{d}z^2$. Wick-rotating the rapidity $\eta \to i\eta_E$ produces the Euclidean polar metric $\mathrm{d}s_E^2 = \rho^2\,\mathrm{d}\eta_E^2 + \mathrm{d}\rho^2 +$ (transverse). Smoothness at $\rho = 0$ (the standard absence-of-conical-defect argument, identical in form to the Hawking–Gibbons argument for the Schwarzschild cigar tip) forces η_E to have period 2π. The accelerated observer's proper time $\tau_p = \eta_E/a$ at fixed $\rho = 1/a$ therefore has period

$$\beta_U \;=\; \frac{2\pi}{a}, \qquad T_U \;=\; \frac{a}{2\pi}, \qquad \text{equivalently} \quad T_U \;=\; \frac{\hbar a}{2\pi c k_B} \text{ (SI units).}$$

M1 identification. The single 2π in β_U is exactly the $k = 0$ volume slot $\mathrm{Vol}(S^1) = 2\pi$ on the angular η_E-circle, identical in structural role to the 2π in Sprint TD Track 4's cigar τ-period and in Track 1's Matsubara S^1_β-circumference. Sprint TD Track 1's `matsubara_spectrum()` called with $\beta = 2\pi/a$ reproduces the Unruh-thermal Matsubara spectrum bit-identically (manual sympy check, residual zero): lowest non-zero bosonic mode equals a (the Unruh analog of Hawking's κ_g surface gravity); lowest fermionic mode equals $a/2 = \pi T_U$; T_U from $1/\beta_U$ identifies bit-exactly with $a/(2\pi)$.

The unified Wick-rotation theorem. The four-witness chain

> *Hartle–Hawking 1976 path-integral → Sewell 1982 generalization to bifurcate Killing horizons → Bisognano–Wichmann 1976 Rindler-wedge modular flow → Unruh 1976 flat-space limit*

is one Wick-rotation theorem with four physical instantiations: the thermal state of an observer with restricted causal access is determined by the regularity period of the Euclidean rotation of their causal-domain coordinates, which equals 2π times $1/(\text{surface gravity})$. Surface gravity values in the four witnesses: $\kappa_g = 1/(4M)$ (Schwarzschild horizon), a (Rindler / Unruh), the boost-rapidity rate (Bisognano–Wichmann), and the Killing surface gravity in the Sewell-generalized statement. In each case, the 2π is the $\mathrm{Vol}(S^1)$ volume factor of the master Mellin engine at $k = 0$ (M1).

Honest scope. Same as § 8.6. The framework's M1 mechanism on $S^1_{\eta_E}$ corresponds, by analogy, to the Wick-rotated modular flow on the Rindler wedge, via the published Wick-rotation prescription (Unruh [25]; Bisognano–Wichmann [21]; Sewell [22]; Hartle–Hawking [24]). This is a *structural correspondence*, not a literal identification: the volume factor on the Riemannian side and the modular-automorphism period on the Lorentzian side are matched by the published Wick-rotation chain, not by an operator-level proof inside the spectral triple. The natural Unruh-side falsifier is structurally the same as Track D's falsifier #1, instantiated at the Rindler wedge instead of the Schwarzschild static exterior: compute the modular Hamiltonian on the truncated metric spectral triple $\mathcal{T}_{n_{\max}}$ for a Rindler-wedge-restricted Camporesi–Higuchi state and verify $\sigma_{i\cdot 2\pi} =$ identity at finite $n_{\max}$. *One falsifier covers both faces* (and also Hawking): the chain that ties the framework 2π to the modular-flow 2π runs through the same KMS-Tomita-Takesaki algebra in every case, so lifting the period-closure check on any one face lifts the structural correspondence on all three. A follow-up beyond sprint scale; priority MEDIUM-HIGH. *Status (2026-09-28):* still open. Paper 42's finite-cutoff period closure posits the generator $K = \mathrm{diag}(2m_j)$ and builds the wedge state from it, so its 2π closure is automatic for integer spectrum and does not run this falsifier.

Cross-references. Paper 32 §VIII Remark `rem:bisognano_wichmann_reading` (Unruh paragraph appended) for the spectral-triple-side statement; Paper 34 §III.15 footnote (extended with Unruh closure of the Track D §5.4 follow-up flag); Paper 34 §V.B Unruh-T off-precision catalogue row (machinery-witness, error class C, a external proper-acceleration input); Paper 34 §VIII Lorentz-boost open question (Track D documentation closure paragraph extended with Unruh four-witness completion).

9 Open questions

1. **Spinor sector (resolved May 2026, sprint KG-5).** Computed. The Dirac-on-S^3 spectrum $|\lambda_n| = n + 3/2$ with degeneracy $g_n = 2(n+1)(n+2)$ (Camporesi–Higuchi [18]) gives the spectral zeta
$$\zeta_{|D|}(s) \;=\; 4\big[\zeta_R(s{-}2,\,3/2) \;-\; \tfrac{1}{4}\zeta_R(s,\,3/2)\big]$$
in the full-Dirac convention (both chiralities, both eigenvalue signs, complex spinor bundle of dimension 4 on S^3). Hurwitz values via Bernoulli polynomials at $a = 3/2$: $B_2(3/2) = 11/12$, $B_4(3/2) = 127/240$. Substitution gives $\zeta_{|D|}(-1) = -17/240$, hence
$$\boxed{E^{\mathrm{Dirac}}_{\mathrm{Cas}} \;=\; -\tfrac{1}{2}\zeta_{|D|}(-1) \;=\; \tfrac{17}{480}} \tag{13}$$
Exact rational, no π, matches the textbook value of Camporesi–Higuchi 1996 (Eq. 5.27), Bytsenko et al. [15] § 4.5, and Dowker–Altaie 1978 (Phys. Rev. D **17**, 417), the vacuum-averaged stress-energy tensor for spinor fields in the Einstein universe, which gives $\langle T_{00}\rangle =$

$17/(1920\pi^2 a^4)$ (a value that paper attributes to Ford); integrated over $\mathrm{Vol}(S^3) = 2\pi^2 a^3$ this is $E = 17/(960a)$, matching the 17/960 single-Weyl-chirality convention line quoted below — the density numeral 1920 and the energy numeral 960 differ by the volume factor and should not be identified. (Its finite-temperature companion, Altaie–Dowker, Phys. Rev. D **18**, 3557, was cited here between 2026-08-28 and this correction; the zero-temperature member of the pair is the right one for a Casimir energy, and the author order differs between them.) Verified to 40 dps numerically. (Attribution corrected 2026-08-28: this sentence previously cited Kennedy–Critchley–Dowker 1980, which is Ann. Phys. **125**, 346 — finite-temperature *scalar* field theory *with boundaries*, neither spinorial nor on the boundaryless S^3. The section number was also inconsistent with this paper's own bibliography entry.) Convention translations (single Weyl chirality with both signs: 17/960; single Majorana real d.o.f.: 17/960; single chirality, single sign: 17/1920) are catalogued in `debug/data/kg5_dirac_casimir.json`.

The per-step transcendental ledger is identical in structure to the scalar case (Section 5): zero π injections in the spatial calculation; the half-integer Hurwitz shift stays in $\mathbb{Q}$ because Bernoulli polynomials have rational coefficients and $3/2 \in \mathbb{Q}$. The Paper 35 prediction is **confirmed in the spinor sector**.

For the temperature-corrected case on $S^3 \times S^1_\beta$, fermions require antiperiodic (Neveu-Schwarz-like) boundary conditions on the temporal circle; the Matsubara modes shift to $\omega_k^t = (2k + 1)\pi/\beta$ instead of $2\pi k/\beta$. π enters at the lowest mode $\omega_0^t = \pi/\beta$ (no zero mode). The injection mechanism is structurally identical to KG-2 — the π is the radian/circumference ratio of the temporal S^1 — shifted by half a Matsubara unit because of the spin structure.

2. **Where does this leave the K combination rule? (Sprint K-CC, clean negative.)** The Paper 2 formula $K = \pi(B + F - \Delta) \approx 1/\alpha$ is the framework's most extreme multi-projection coincidence. Under Prediction 1, the explicit π on the right-hand side flagged two candidate readings: (i) K is, despite appearances, a single-projection statement of the spectral-action / temporal-compactification kind (cf. the Cardano root $\Lambda_\infty = 3.7102\ldots$ from the SD two-term exactness theorem; the π would then be the $k = 0$ volume slot $\mathrm{Vol}(S^2)/4$, identified in Sprint A), or (ii) the prediction is false in the multi-loop / infinite-Dirichlet sector, and the π has a non-temporal source that the framework has not yet named. Sprint K-CC (May 2026) tested possibility (i) along three independent axes with a clean triple negative:

   (a) PSLQ at 100 dps against a 244-element basis of framework invariants (B, F, Δ, κ, $|\lambda_n|$, g_n, half-integer Hurwitz values, Bernoulli at half-integers, S^3 Seeley–DeWitt coefficients, π powers, and small-exponent products) returned **no identification of** Λ_∞. The depressed cubic $2\Lambda^3 - \Lambda - 4K/(\pi\sqrt{\pi}) = 0$ is recovered as positive control; removing K from the basis kills the identification entirely.

   (b) Search for a natural selection criterion for Λ from S^3 spectral data (35 cutoff–target pairs, 37 composite-eigenvalue candidates, plus critical-point analysis) returned only tautological "hits" (Gaussian trace at Λ_∞ literally equals K/π by construction). The cleanest non-tautological near-miss — the Gaussian trace equating to $B + F$ — gives Λ at relative distance 1.86×10^{-4} from Λ_∞, ∼three orders of magnitude looser than the K residual itself (4.8×10^{-7} for the plain sum; 8.8×10^{-8} via Paper 2's self-consistency root $1/\alpha + \alpha^2 = K$), and so cannot explain the K precision even if elevated.

   (c) The unified heat-kernel approach is blocked by **Theorem T9** (Paper 28): $\zeta_{D^2}(s)$ at integer s lies in the pure-Tate ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ (the $\sqrt{\pi}$ slot is empty at integer s). Symbolic computation gives $\zeta_{D^2}(2) = \pi^2 - \pi^4/12$ and $\zeta_{D^2}(4) = \pi^6(168 - 17\pi^2)/1260$. $F = \pi^2/6$ does not appear at any integer s of the unified expansion. Independently of

PSLQ, this is an *algebraic obstruction* to the unified-CC reading: B, F, Δ live in three categorically different spectral objects on two different bundles (finite scalar Casimir trace at $m = 3$; infinite scalar Fock–Dirichlet sum at $s = d_{\max}$; single-level Dirac mode degeneracy at $n = 3$ on the spinor bundle), and no single CC heat-kernel expansion contains all three as different terms.

Reading (i) is therefore ruled out at the framework's current resolution. The remaining live readings are (ii) revised: the π in K has a non-temporal source the framework has not yet named, or (iii) K is a coincidence below the framework's intrinsic resolution and Prediction 1 is consistent but non-predictive at the level of the observed agreement (K to 4.8×10^{-7}; the cubic's root to 8.8×10^{-8}). The K-CC sprint independently verified the closed form $\zeta_{D^2}(2) = \pi^2 - \pi^4/12$ as a new Paper 28 identity (consistent with $T9$, not previously logged). Sprint provenance: `debug/kcc_sprint_memo.md`.

3. α^5 **Lamb shift residual: scoped, reframed, and confirmed.** Sprint LS-5 (May 2026) scoped the two-loop QED calculation on the framework and found two distinct results, both since confirmed by sprint LS-6a. *First*, the framework is structurally ready for two-loop QED on S^3: the mode-count budget is favorable (∼2,870 terms at $n_{\max} = 20$ from SO(4) selection rules), and the infrastructure pieces exist (`qed_two_loop.py`, `qed_self_energy.py`, the LS-3/LS-4 bound-state Sturmian projection); the work is additive rather than blocked, with an estimate of ∼8 follow-on sprints to close the multi-loop ceiling. *Second*, the original test target was misidentified. The genuine two-loop α^5 multi-loop contribution to the $2S$ Lamb shift is approximately +7.10 MHz (Eides–Grotch–Shelyuto 2001 Table 7.4 cumulative central value), not the $\sim +29$ MHz LS-3 residual. Sprint LS-6a (May 2026) confirmed the reframing: re-deriving the same one-loop physics in the Eides § 3.2 canonical convention identified that the original LS-1 implementation had inadvertently subtracted the Uehling kernel constant $4/15 = (10/9 - 38/45)$ from the canonical $10/9$ self-energy coefficient (Uehling already enters separately as a VP contribution). The convention shift from LS-1 to LS-6a is exactly +27.13 MHz at $n = 2$, $Z = 1$, structurally identifiable as $(4/15) \cdot \alpha^3 Z^4/(\pi n^3) \cdot (\text{Ha-to-MHz})$. LS-6a predicts 1052.19 MHz, with residual −5.65 MHz / − 0.534% against experimental 1057.85 MHz — a 5.8× reduction in absolute error vs the LS-1 one-loop ceiling (−32.78 MHz); vs LS-3 (−28.85 MHz) the reduction is 5.1×. Sprint LS-7 (May 2026) made a first pass at the two-loop SE on Dirac-S^3 via iterated CC spectral action and returned two findings. *Structural confirmation*: the two-loop SE prefactor $(\alpha/\pi)^2(Z\alpha)^4/n^3$ emerges with the $1/\pi^2$ tracing to two iterated Schwinger proper-time integrations, exactly as Prediction 1 specifies; weak test because the same prefactor appears in flat-space two-loop QED. *Critical residual reframing*: the −5.65 MHz LS-6a residual is *not primarily multi-loop QED*. The Eides–Grotch–Shelyuto 2001 Table 7.3 (α^5 multi-loop QED proper) totals only $\sim +1.20$ MHz; the remaining $\sim +4.4$ MHz of the residual must be non-loop physics; the itemized non-loop entries are recoil −2.40 MHz, finite nuclear size +0.138 MHz (the 2S value; the retired +1.18 is the 1S shift), hyperfine averaging $\sim +5.0$ MHz. These three sum to $\sim +2.74$ MHz; adding the $\sim +1.20$ MHz multi-loop term gives $\sim +3.94$ MHz for the four itemized contributions against the −5.65 MHz residual — $\sim 70\%$ closure, not a match to within the literature precision. (Corrected 2026-08-29: an earlier correction note printed +3.78/+4.98 — sums still computed with the retired +1.18 MHz finite-size entry; the +0.138 MHz 2S value gives +2.74/+3.94, in agreement with Paper 34's Lamb autopsy. Cross-check: $5.65 - 3.94 = 1.71$ MHz reproduces Paper 34's tabulated +1.72 MHz Lamb-autopsy residual — the unattributed remainder IS the autopsy residual.) The earlier LS-5 / LS-6a memos had used Eides Table 7.4 (cumulative α^5 contributions of all kinds) as the multi-loop reference; this was a misreading. Implica-

tions: (i) the framework's one-loop closure is even cleaner than originally framed — only $\sim +1.20$ MHz of the residual is missing-multi-loop physics; (ii) the genuine multi-loop test target for Prediction 1 is sharpened to $\sim +1.20$ MHz, not $\sim +5.65$ MHz; (iii) the strong test (sprint LS-8a) is the native derivation of the dimensionless bracket coefficient $C_{2S} = +3.63$ from iterated CC spectral action plus bound-state Sturmian projection, with the literature bracket value used as a check rather than as input. The remaining $\sim +4.4$ MHz of non-loop physics is within scope for the rest-mass projection (Paper 34 § III.14) and Paper 23's nuclear-electronic composed Hamiltonian, but is not itself a test of Prediction 1.

Sprint LS-8a (May 2026) closed the strong-test attempt at C_{2S} extraction with a clean structural verdict. The bare iterated Connes–Chamseddine spectral action on Dirac-S^3 (rainbow plus crossed topologies, full SO(4) vertex selection at four vertices, bound-state Sturmian projection at $n_{\text{ext}} = 1$ for hydrogen $2S$) faithfully reproduces the UV-divergent *integrand* of two-loop QED: the $(\alpha/\pi)^2(Z\alpha)^4/n^3$ prefactor emerges as predicted, the sign is correct at every $n_{\max}$, and the sum diverges as raw $\sim 6.6 \cdot N^{3.43}$—power-law UV divergence inherited from flat-space QED. Two natural regularizations within the GeoVac toolkit fail: subtracting the disconnected $[\Sigma^{1L}]^2$ piece removes <0.1% of the total at every $n_{\max}$ (the divergence sits in the genuine connected diagram), and Drake–Swainson asymptotic subtraction (Paper 34's 13th projection) does not apply to power-law divergences. Finite extraction of C_{2S} requires Z_2 and δm renormalization counterterms which are *not generated* by the bare CC spectral action. This places a clean scope boundary between one-loop closure (achieved at sub-percent in Paper 36 via Drake–Swainson plus the Eides § 3.2 canonical convention) and two-loop closure (requires renormalization machinery the framework does not natively produce). **Refined Prediction 1:** the GeoVac framework controls the π content of the *finite parts* of any QED observable; UV divergences of multi-loop contributions are inherited from the underlying field theory and renormalized by counterterms that the framework does not autonomously generate. This is consistent with and strengthens the original Prediction 1: π enters via temporal integration; renormalization is a separate projection not yet in the Paper 34 catalogue. Sprint provenance: `debug/ls5_two_loop_scoping_memo.md`, `debug/ls6a_eides_convention_memo.md`, `debug/ls7_two_loop_se_memo.md`, and `debug/ls8a_two_loop_sel`

4. **Compactification beyond S^1_β.** We tested only the $S^3 \times S^1_\beta$ compactification. More general temporal identifications (with twists, with curvature, with multiple temporal directions) would give more general π-injection patterns. Whether the prediction (Section 7) extends to all such compactifications, or only to the standard Matsubara case, is open.

10 Conclusion

The bare relativistic Klein-Gordon spectrum on $S^3 \times \mathbb{R}$ is π-free in the algebraic-extension ring $\mathbb{Q}[\sqrt{d_1}, \sqrt{d_2}, \ldots]$ for every rational mass-squared. π enters the spectrum at exactly one step – temporal compactification – through the Matsubara frequency $2\pi k/\beta$. The conformally coupled massless scalar Casimir energy on unit S^3 is the exact rational $1/240$; the Stefan-Boltzmann $\pi^2/90$ enters only through the Matsubara sum. The structural reading is that mass is a parameter projection that preserves the algebraic-extension ring of the bare graph; the observation / temporal-window projection is the categorically distinct projection that introduces π. These should be split as separate rows in Paper 34's projection dictionary.

The framework-wide prediction is that a GeoVac observable contains π if and only if its evaluation includes a continuous integration over a temporal or spectral parameter promoted from the discrete graph spectrum. Existing GeoVac observables uniformly confirm this prediction: Pauli

counts, magic numbers, hydrogenic eigenvalues, Bargmann–Segal lattice, and graph-native QED matrix elements are all π-free; vacuum polarization, β-function, and Stefan-Boltzmann all contain π exactly because their evaluation includes a continuous Schwinger-proper-time or Matsubara integration.

With the addition of Sprint TX-B's five-observable falsification panel (§ 7.3), the prediction has been tested on a cumulative 208/208 individual checks across the KG and TX-B sprints with no exceptions, and graduates from falsifiable conjecture to load-bearing principle of the framework (Remark 4). TX-B also produced a new clean closed form: the spatial Casimir of the 3D HO on the Bargmann–Segal S^5 lattice is $E_{\mathrm{Cas}}^{\mathrm{HO},S^5} = -17/3840$ (exact rational), structurally a sibling of the scalar S^3 Casimir (1/240, KG-3) and the Dirac S^3 Casimir (17/480, KG-5). The shared mechanism is half-integer Hurwitz at shift $a = 3/2$, with Bernoulli polynomials at 3/2 rational — the universal driver of rationality across all three Casimir sectors. The Coulomb/HO/Dirac asymmetry of Paper 24 (operator-order, calibration π, Wilson gauge with matter) does not extend to the Casimir transcendental class.

The sharpest open question this paper does not resolve is the Paper 2 $K = \pi(B+F-\Delta) \approx 1/\alpha$ coincidence: the explicit π on the right-hand side is a candidate counter-example to the prediction, but only if the projections that produce K are genuinely discrete-graph quantities and not single-projection spectral-action-type statements in disguise. Sprint K-CC has since *ruled out* the single-projection reading at the framework's current resolution (no single Chamseddine–Connes heat-kernel expansion contains B, F and Δ as terms); the live readings are the revised multi-projection one and the coincidence one, per WH5. The open step is therefore to discriminate *those two*, a structural follow-on to Sprint A (2026-04-18) and the WH1 Marcolli–van Suijlekom upgrade. (Updated 2026-08-28: the earlier wording proposed as a next step a question this paper's own Section 9 reports closed.)

References

[1] GeoVac Project, "Angular Momentum as Information Geometry: Isotropic Packing and the Recurring Angular Cross-Section," Paper 0 (2026).

[2] GeoVac Project, "The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice," Paper 7 (2025).

[3] GeoVac Project, "Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory," Paper 14 (2025).

[4] GeoVac Project, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," Paper 18 (2026).

[5] GeoVac Project, "Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems," Paper 22 (2026).

[6] GeoVac Project, "Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice," Paper 23 (2026).

[7] GeoVac Project, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," Paper 24 (2026).

[8] GeoVac Project, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," Paper 28 (2026).

[9] GeoVac Project, "The Universal/Coulomb Partition of the GeoVac Spectral Triple," Paper 31 (2026).

[10] GeoVac Project, "The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage," Paper 32 (2026).

[11] GeoVac Project, "The 1+6+1 Partition of QED Selection Rules on S^3," Paper 33 (2026).

[12] GeoVac Project, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," Paper 34 (2026).

[13] GeoVac Project, "Sprint KG Memo: Klein-Gordon on $S^3 \times \mathbb{R}$ – where does π enter?" `debug/kg_sprint_memo.md` (May 2026).

[14] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space*, Cambridge University Press, 1982.

[15] A. A. Bytsenko, G. Cognola, E. Elizalde, V. Moretti, and S. Zerbini, *Analytic Aspects of Quantum Fields*, World Scientific, 2003 (§ 4.5).

[16] J. S. Dowker and R. Critchley, "Effective Lagrangian and energy momentum tensor in de Sitter space," *Phys. Rev. D* **13**, 3224 (1976).

[17] L. H. Ford, "Quantum Vacuum Energy in General Relativity," *Phys. Rev. D* **11**, 3370 (1975).

[18] R. Camporesi and A. Higuchi, "On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces," *J. Geom. Phys.* **20**, 1 (1996).

[19] V. Morinelli, "A geometric perspective on Algebraic Quantum Field Theory," arXiv:2412.20410 (2024).

[20] J. E. Tener, "The Bisognano-Wichmann property for non-unitary Wightman conformal field theories," arXiv:2506.10625 (2025).

[21] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* **17**, 303–321 (1976).

[22] G. L. Sewell, "Quantum fields on manifolds: PCT and gravitationally induced thermal states," *Annals of Physics* **141**, 201–224 (1982).

[23] B. S. Kay and R. M. Wald, "Theorems on the uniqueness and thermal properties of stationary, nonsingular, quasifree states on spacetimes with a bifurcate Killing horizon," *Physics Reports* **207**, 49–136 (1991).

[24] J. B. Hartle and S. W. Hawking, "Path-integral derivation of black-hole radiance," *Phys. Rev. D* **13**, 2188–2203 (1976).

[25] W. G. Unruh, "Notes on black-hole evaporation," *Phys. Rev. D* **14**, 870–892 (1976).